Multi-Objective Reinforcement Learning for Autonomous Driving and Adaptive Sensor Optimization

Zachary L. Johnson-Scott
University of Tennessee – Knoxville
ECE 517
Final Project

Problem Statement:

SAE Level 5 autonomous vehicles are estimated to require approximately 800–1,200 watts of power for autonomous driving systems, contributing to reduced vehicle range, increased electrical system complexity, greater thermal management demands, and higher associated carbon emissions (Sudhakar et al., 2023). These vehicles rely on multiple sensing technologies, many of which operate continuously to maintain situational awareness. Multisensory perception systems are estimated to require between 25 and 39 sensors (Viktor & Kiss, 2026).

However, the continuous operation of all sensors may not be necessary across every driving condition. Different scenarios present varying levels of complexity, uncertainty, and risk, potentially allowing sensor utilization to be dynamically adjusted without compromising safety. This raises a fundamental research question: Can a reinforcement learning agent learn to anticipate sensing requirements and dynamically optimize sensor activation across changing driving conditions while maintaining safe vehicle operation and minimizing energy consumption?

This project investigates the feasibility of using multi-objective reinforcement learning to jointly optimize autonomous driving performance and adaptive sensor utilization, balancing safety, situational awareness, and energy efficiency.

Motivation:

This project connects reinforcement learning with my broader research in event-based cameras, neuromorphic computing, and energy-efficient autonomous systems. Traditional autonomous vehicles continuously operate multiple computationally expensive sensors, potentially consuming energy even when additional sensing is unnecessary.

I'm interested in exploring whether an RL agent can learn when additional sensory information is needed, rather than continuously operating every sensor. This could provide insights into more energy-efficient autonomous systems and establish a foundation for future experimentation using realistic sensor simulations in CARLA and eventually physical event-camera hardware.

Objective:

Design and simulate a multi-sensor environment to evaluate a multi-objective reinforcement learning framework that can dynamically optimize sensor activation and deactivation for autonomous vehicle operations across varying traffic conditions. With a final goal of deploying a

Multi-Objective Reinforcement Learning for Autonomous Driving and Adaptive Sensor Optimization

RL policy that can minimizing energy consumption while maintaining vehicle safety and overall driving performance.

Environment Design:

The proposed environment will be developed by extending HighwayEnv, leveraging existing vehicle dynamics, traffic simulation, and kinematic observations. The environment will be customized to incorporate multiple simulated sensors, each providing different types of observational information, sensing capabilities, and energy requirements. Will implement a custom sensor simulation layer, joint action space and reward model.

The agent will operate under partial observability, with the information available dynamically determined by sensor activation states. A fixed observation structure will accommodate sensor availability, measurement uncertainty, and potentially outdated information.

The action space will combine HighwayEnv's existing driving actions, such as acceleration, deceleration, and lane changes, with additional actions allowing the agent to activate or deactivate individual sensors. Each sensor will have an associated energy and computational cost, along with a modeled information benefit. Additional costs associated with downstream perception processing may also be incorporated.

The reward function will consider multiple objectives, including collision avoidance, driving performance, task completion, and energy consumption. Through reinforcement learning, the agent will develop a policy that balances safe highway navigation with efficient sensor utilization across varying traffic conditions. Collison will not terminate the run as it might encourage the system to collide to reduce energy used.

Reinforcement Learning Approach:

The project will initially implement a Q-learning framework using a weighted multi-objective reward function to balance three competing objectives: driving safety, driving performance, and sensor energy efficiency. Each objective will have an independently calculated reward component, with adjustable weights determining its relative importance during training.

Multiple weighting configurations will be evaluated to examine tradeoffs between objectives. Pareto-front analysis will be used to identify nondominated policies, with predefined safety thresholds helping determine which solutions are acceptable. Vector-valued Q-learning will also be explored as a potential method for maintaining separate action-value estimates for each objective.

Once baseline performance is established, the framework will be extended toward a Deep Q-Network (DQN) architecture to accommodate more complex observations and decision-making scenarios.

Multi-Objective Reinforcement Learning for Autonomous Driving and Adaptive Sensor Optimization

Technical Challenges:

1. Multi-Objective Optimization: The agent must balance competing objectives, including collision avoidance, driving efficiency, and sensor energy consumption. Optimizing one objective may negatively affect another.
2. Designing sensor observation and energy cost is an unknown as the perception will be sudo, thus an approximation must be realistic but work within the environment limitations. Hence design of the sensor information provided, and compute cost is paramount to success.
3. Partial Observability and Sensor Uncertainty: The agent will have limited information about surrounding traffic. Activating additional sensors improves situational awareness but increases energy consumption. The agent must learn when acquiring additional information is worth the cost.
4. Dynamic Traffic Conditions: Randomized vehicle speeds, sudden braking, and lane changes will require the agent to adapt its driving and sensing strategies to unpredictable situations.

Research Question and Evaluation:

Can a multi-objective reinforcement learning framework simultaneously optimize automated safe driving decisions and sensor related energy consumption? How do different policies affect the tradeoffs between safety, driving performance and sensor related energy consumption?

Several examples of evaluation below, these aren't all inclusive but demonstrate evaluation approach. The agent will be evaluated across three primary objectives: driving safety, driving performance, and sensor energy efficiency. Performance metrics will include collision rates, minimum time-to-collision, distance traveled, successful episode completion, and estimated sensor energy consumption per kilometer. Evaluation will be conducted across randomized traffic scenarios, including varying traffic densities, sudden braking, and lane changes, to assess policy robustness and generalization. Policies will be assessed against predefined safety and driving performance thresholds to determine whether energy savings can be achieved without unacceptable degradation in vehicle operation.

GOAL	METRIC	DESIRED OUTCOME
Safety	Collision rate	Maintain or improve safety
Performance	Distance traveled and or average speed, average rate of change in speed (less is better for smooth driving)	Efficient highway navigation
Energy	Average number of sensors activated per	Reduce energy consumed

	second, modeled sensor energy per episode		
Policy	**Driving Strategy**	**Sensor Strategy**	**Purpose**
Baseline	Best Recorded	Always On	Establish Baseline
A	Slow but Safe	Adaptive 80/20	Trade Off
B	Fast and Less Safe	Adaptive 50/50	Trade Off
C	Medium and Mostly Safe	Adaptive 20/80	Trade Off

Expected Results:

The reinforcement learning agent is expected to learn optimal state-action pairs that minimize sensor utilization/energy consumption by dynamically selecting the most appropriate sensors for varying driving conditions, while maintaining at least 70% of the driving performance achieved by the baseline configuration with all sensors continuously active.

Project Plan / Timeline:

Estimated timeline will try to develop faster including starting environment development this week with basic HighWanyEnv implementations.

WEEK	TASK TO COMPLETE
1	Observation Space and Sensor Research, Reward Experimentation, Custom HighWayEnv / Testing
2	Baseline Implementation / Q - > DQN -> Other
3	Full Sensor Integration, Observation and Action Space Changes as needed, Reward tuning, MORL
4	Pareto Front, Evals, Final Result, Presentation

References

Farama Foundation. (n.d.). Make your own custom environment. Gymnasium Documentation. https://gymnasium.farama.org/main/introduction/create_custom_env/

Leurent, E. (2018). An environment for autonomous driving decision-making [Computer software]. GitHub. https://github.com/Farama-Foundation/HighwayEnv

Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A. A., Veness, J., Bellemare, M. G., Graves, A., Riedmiller, M., Fidjeland, A. K., Ostrovski, G., Petersen, S., Beattie, C., Sadik, A., Antonoglou, I., King, H., Kumaran, D., Wierstra, D., Legg, S., & Hassabis, D. (2015). Human-level control through deep reinforcement learning. Nature, 518, 529–533. https://doi.org/10.1038/nature14236

Sudhakar, S., Sze, V., & Karaman, S. (2023). Data centers on wheels: Emissions from computing onboard autonomous vehicles. IEEE Micro, 43(1), 29–39. https://doi.org/10.1109/MM.2022.3219803

Van Moffaert, K., & Nowé, A. (2014). Multi-objective reinforcement learning using sets of Pareto dominating policies. Journal of Machine Learning Research, 15(107), 3663–3692. https://jmlr.org/papers/v15/vanmoffaert14a.html

Viktor, P., & Kiss, G. (2026). Sensors in self-driving vehicles: A detailed literature review and new trends. Sensors, 26(7), Article 2153. https://doi.org/10.3390/s26072153

AI Use Disclosure

OpenAI's ChatGPT 6 and Grammarly were used to assist with grammar, clarity, sentence structure, and organization during the editing process. AI assistance was limited to refining written content, while the research concepts, project design, and technical decisions were developed by the author.